Venkata Lakshmi Parimala Pasupuleti

Software Engineer, AI/ML · LLM-Powered Services · AI Orchestration · MCP · Production AI Platform · Python · Java · AWS · Event-Driven

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github.com/venkatalakshmiparimala | Fairfax, Virginia, USA (Open to Relocation)

SUMMARY

Software engineer with **7+ years of experience** building production systems and AI-powered services. Currently at Intuit: **LLM-powered workflows** integrating Anthropic Claude into production financial services via tool calling, event-driven pipelines, structured outputs, and CI eval harnesses. Built **Leasel** - a production AI platform: 8-agent LangGraph orchestration, **Model Context Protocol (MCP)** server integration, OCR-grounded document understanding, hybrid-search RAG, and a **CI eval harness** (93.6% precision, 1,240 test cases). Built **AgentProbe** - an open-source **AI orchestration observability platform** with cost-aware multi-provider routing, CI regression gating, and per-run tracing. **7+ years** in production: scalable microservices, async event-driven processing (RabbitMQ), PHI-compliant AWS systems at Entrata (100K+ users). Strong in **Python, Java, FastAPI, AWS**, and full production lifecycle - prototype through production. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

LLM-Powered Services · Generative AI Features · AI Orchestration Patterns
Model Context Protocol (MCP) · Tool Calling · Multi-Agent Orchestration
Scalable APIs · Microservices · Event-Driven Workflows · Async Processing
Document Understanding · OCR · Multimodal Embeddings · RAG
Prototype → Production · Eval Harnesses · Measure Outcomes · CI Regression
Reliability · Resilience · Security · Cost Efficiency · Production Support
Python · Java · FastAPI · Spring Boot · LangGraph · LangChain
AWS (S3 · Lambda · SQS · API Gateway · CloudWatch · IAM)
PostgreSQL · ChromaDB · Redis · Docker · Kubernetes · CI/CD
Guardrails · HITL · Input Validation · Structured Outputs · Observability
Business Workflow Automation · Data Pipelines · Enterprise Integrations
React · TypeScript · REST APIs · GraphQL · Agile · Cross-Functional

EXPERIENCE

Software Engineer

Jun 2026 - Present

Intuit — Virginia, USA (Remote)

Financial services AI - LLM-powered products for 100M+ customers

- **Built LLM-powered services and reusable components** in production: integrated Anthropic Claude into Python/FastAPI services via tool calling, structured outputs, and retrieval-grounded context - connecting AI to customer financial data, internal services, and business workflows. Iterated rapidly from prototype to production on multiple AI features.
- **Designed and built an LLM eval harness** with CI regression gating (golden datasets, LLM-as-judge scoring) - enabling fast, confident iteration on AI quality; structured logging + CloudWatch instrumentation for full production observability. Implemented guardrails: input validation, sensitive-data handling, fallbacks, and HITL for safety-critical paths.
- Build ML data pipelines (scikit-learn, XGBoost); React/TypeScript front ends; deploy via Docker + GitHub Actions CI/CD to AWS with asynchronous processing and event-driven patterns.

Senior Software Engineer

Oct 2020 - Jul 2024

Entrata — Affordable Housing SaaS

Regulated multi-tenant platform · 100K+ monthly active users · Python, Java, PHP, AWS

- **Owned scalable microservices and async event-driven workflows** end-to-end: RabbitMQ-based processing for rent updates across thousands of tenant records (idempotency guarantees, retry logic, dead-letter routing); PHI-compliant AWS S3 migration (Libsodium encryption, IAM least-privilege, 100% coverage, zero incidents); HUD compliance ETL (180K records, zero violations across 6 years).
- Led Docker **microservices redesign** (50% scalability improvement, 30% faster deployments); optimized 200+ SQL queries (30% performance gain); Python/Bash automation eliminating 80% of manual operational work; mentored 5+ engineers across 120+ Agile sprints.

Junior Software Engineer

Sep 2018 - Oct 2020

Entrata India — Pune

- Built backend modules (Python, PHP, Java, PostgreSQL); delivered large-scale data integration workflows (1.6M records); led AWS S3 migration from legacy storage.

PROJECTS

Leasel - Production AI Platform: MCP · AI Orchestration · Document Understanding · Eval

2026 | Python, LangGraph, FastAPI, ChromaDB, MCP, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production **LLM-powered platform** automating multi-step document workflows: **8-agent LangGraph orchestration** with tool calling, conditional routing, and reusable components; **Model Context Protocol (MCP) server integration** for shared tool access across agents; **document understanding** via OCR ingestion, vector search, semantic reranking, and citation grounding; HITL guardrails for regulated decisions; **CI eval harness** (LLM-as-judge, golden datasets: 93.6% precision, 1,240 test cases). Iterated from prototype through 3 production surfaces with full observability.

AgentProbe - AI Orchestration Observability & Multi-Provider Eval Platform

2025 | Python, FastAPI, PostgreSQL, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Open-source platform for **production AI reliability**: offline golden-dataset evals, LLM-as-judge scoring, CI regression gating, and structured failure classification; **cost-aware model routing** across Anthropic, OpenAI, Azure OpenAI, AWS Bedrock, Groq, and Ollama; full per-run tracing and instrumentation for AI behavior profiling. Used as a **production release gate** before every AI deployment.

AI on FHIR — Document & API Integration with Multimodal Embeddings

2024 | Python, Flask, React, BERT, LLMs, FHIR REST APIs, Docker | github.com/venkatalakshmiparimala/AI_on_FHIR

- Built a **generative AI feature** connecting natural-language questions to FHIR REST API calls using LLMs, BERT, and **multimodal embeddings** - cut clinical lookup time ~60%. Demonstrates end-to-end pattern of **connecting AI models to business data and internal services**: document understanding, OAuth-secured API integration, HIPAA-style RBAC, Docker-containerized.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia | GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017